Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

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Middle East Technical University

current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

Why this lecture?

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

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- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

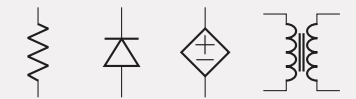
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **Substitution theorem** — replace a part of a circuit by a source
3. **One-ports and equivalence** — the same i–v characteristic at the port
4. **Series and parallel** — combine resistors and sources
5. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

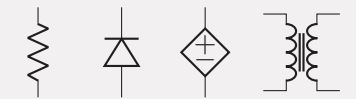


later: the ideal op-amp

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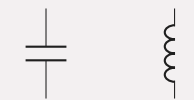
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Its law has a derivative.

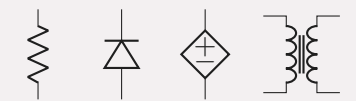


also coupled inductors

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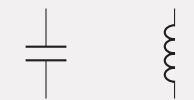
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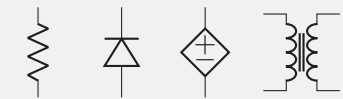
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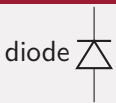
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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

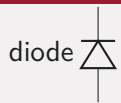


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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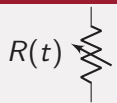
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Time-invariant?

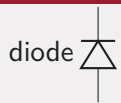


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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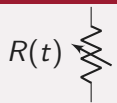
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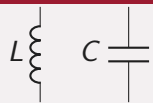
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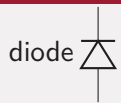


Dynamic if it contains at least one dynamic element: an L or a C.

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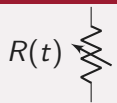
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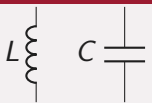
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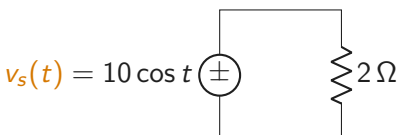
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LTI resistive circuit = linear + time-invariant + resistive:

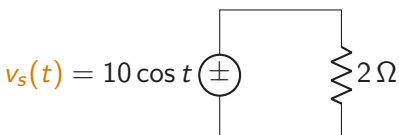
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

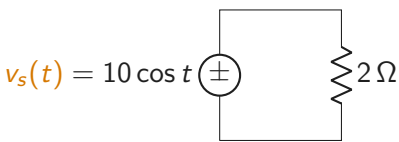
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The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

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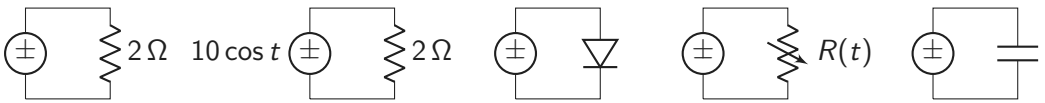
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Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

Which circuit is which?

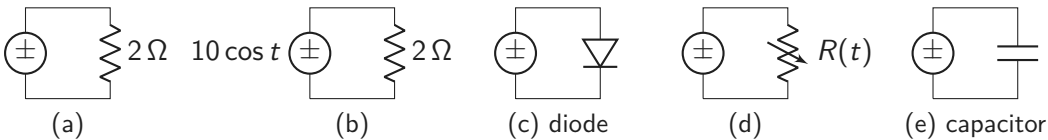


(a) (b) (c) diode (d) (e) capacitor

Sources: $5\,\mathrm{V}$ except (b). In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

Which circuit is which?



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	(a)	(b)	(c)	(d)	(e)
linear	✓	✓	×	✓	✓
time-invariant	✓	✓	✓	×	✓
resistive	✓	✓	✓	✓	×
LTI resistive	✓	✓	×	×	×

What we gain

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$$v = Ri \quad \text{(resistor)}, \qquad v = \mu v_x, \quad i = g v_x, \quad \ldots \quad \text{(dependent sources)}$$

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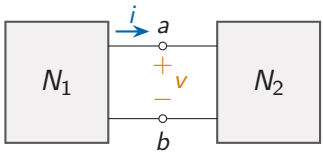
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- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

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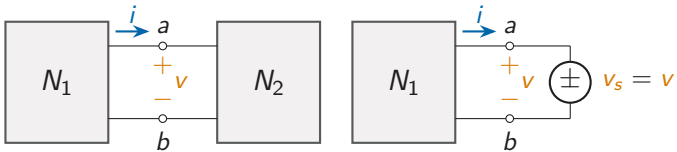
The substitution theorem



(I)

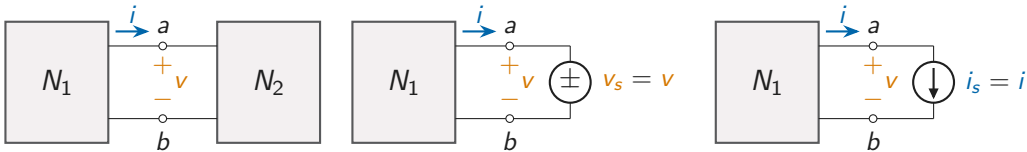
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



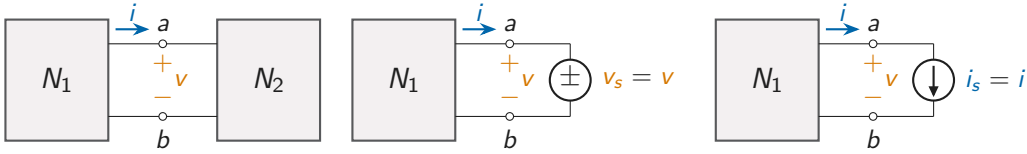
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



... or by a current source with $i_s(t) = i(t)$.

The substitution theorem



Substitution theorem

No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

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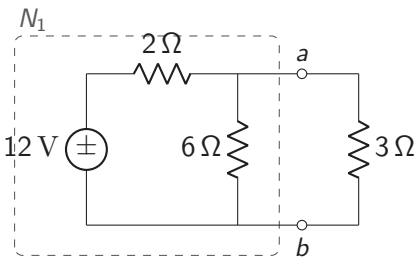
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One condition

The new circuit must have a unique solution.

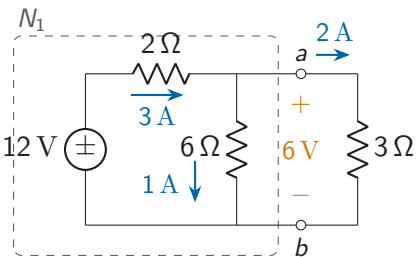
Example: N_1 a single current source and N_2 replaced by a current source $\Rightarrow$ v undetermined.

Example: substitution



N_2 is the 3 Ω resistor. Find v, i and the currents inside N_1.

Example: substitution

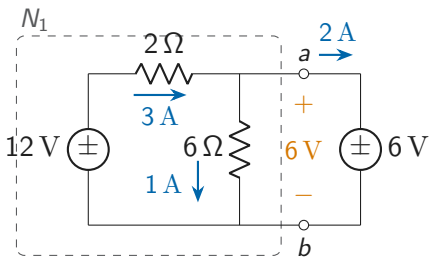


(I) $6\,\Omega \parallel 3\,\Omega = 2\,\Omega$; the two $2\,\Omega$ share $12\,\mathrm{V}$:

$v = 6\,\mathrm{V}, \quad i = 6/3 = 2\,\mathrm{A}$

Inside N_1: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

Example: substitution



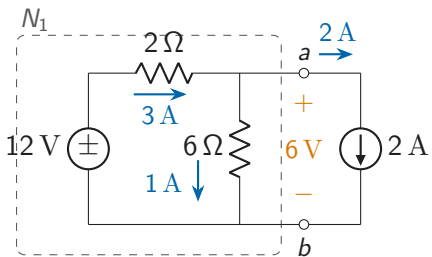
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(II) A $6\,\mathrm{V}$ source fixes $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

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(III) A $2\,\mathrm{A}$ source: KCL at a, $\frac{12-v}{2} = \frac{v}{6} + 2$, so $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

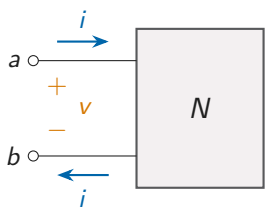
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Ports and one-ports

Port, one-port

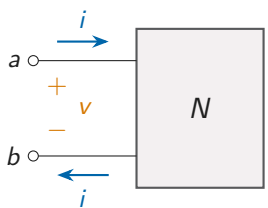
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



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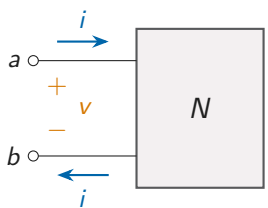


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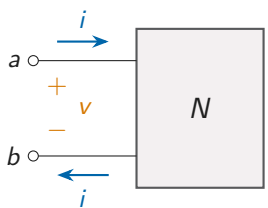


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.

Ports and one-ports

Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

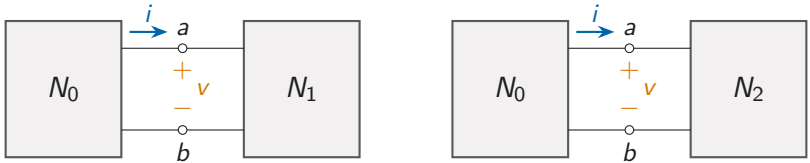


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

Equivalent one-ports

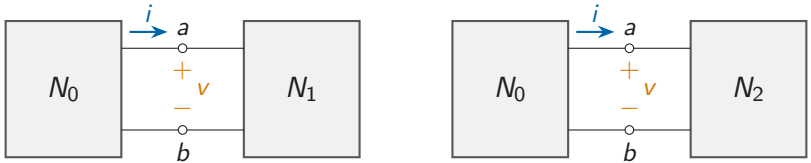
$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



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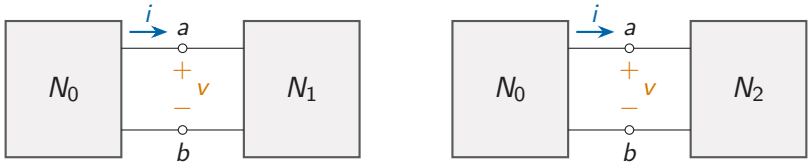


For any N_0: every branch voltage and current inside N_0 is the same in both.

Equivalent one-ports

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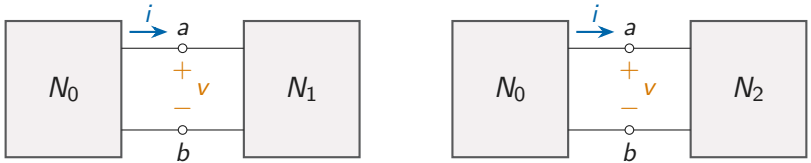
For any N_0: every branch voltage and current inside N_0 is the same in both.

Why: the equations are those of N_0 plus the i–v characteristic at the port.

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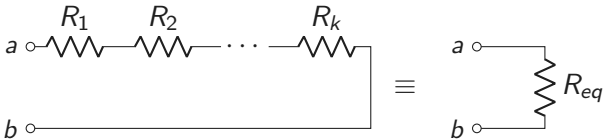
Inside N_1 and N_2 the branch values may differ.

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **Substitution theorem** — replace a part of a circuit by a source ✓
3. **One-ports and equivalence** — the same i–v characteristic at the port ✓
4. **Series and parallel** — combine resistors and sources
5. **One source and one resistor** — exchange a voltage source and a current source

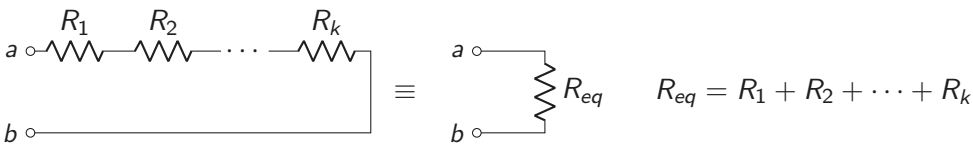
Series connection

Same current i through every element (KCL); the voltages add (KVL).



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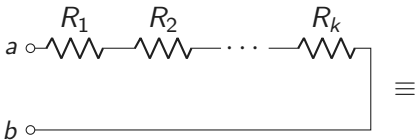


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

Series connection

Same current i through every element (KCL); the voltages add (KVL).

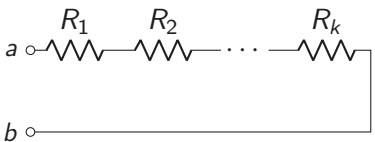


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

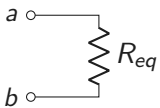
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

Series connection

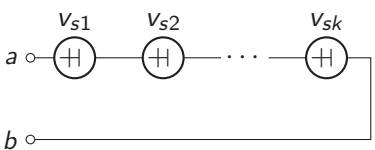
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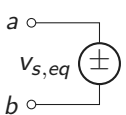
$\equiv$



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$



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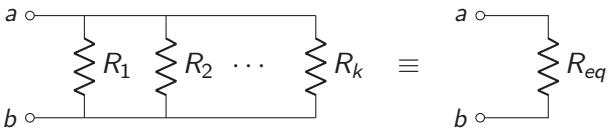


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

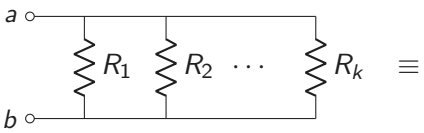
Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



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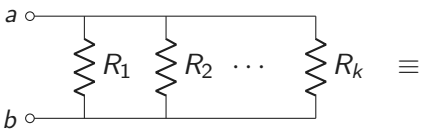


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

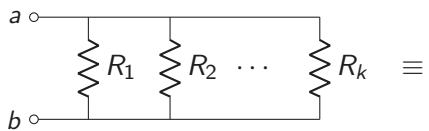


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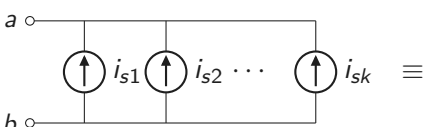
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

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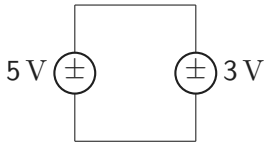
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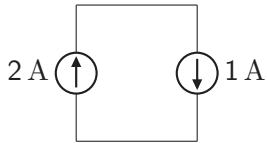
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All arrows towards a; a source turned the other way enters with $-$.

Two connections to avoid



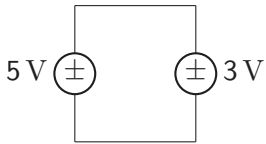
two voltage sources in parallel



two current sources in series

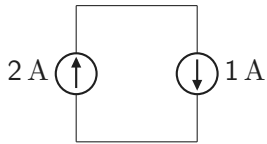
Are these allowed?

Two connections to avoid



two voltage sources in parallel

KVL fails: $5 \neq 3$.



two current sources in series

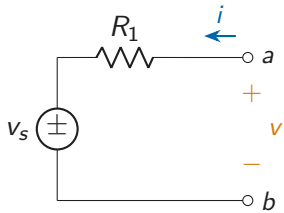
KCL fails: $2 \neq 1$.

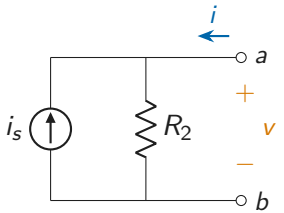
With equal values they are allowed, but the shared current (or voltage) is not fixed.

Roadmap

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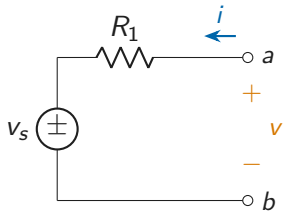
One source and one resistor

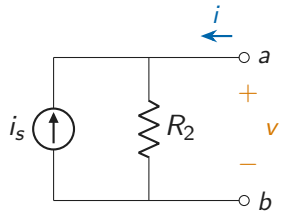




When do these two one-ports have the same i–v characteristic?

One source and one resistor

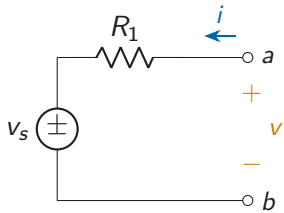




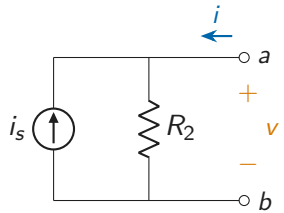
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



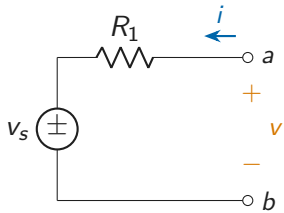
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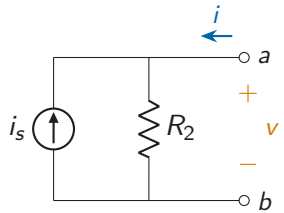
$v = R_2(i + i_s) = R_2 i + R_2 i_s$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

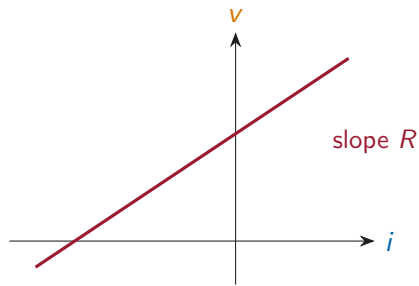
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



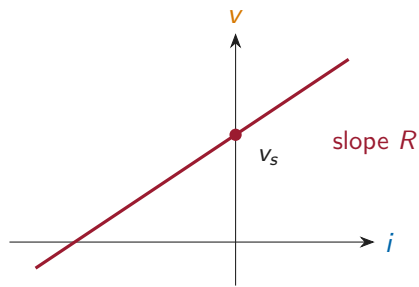
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Open circuit ($i = 0$): $v = v_s$.



Source transformation

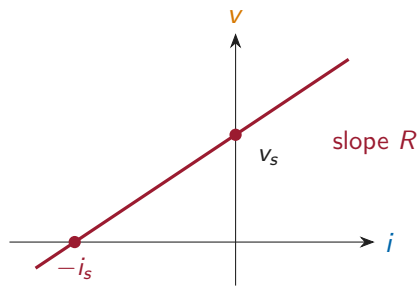
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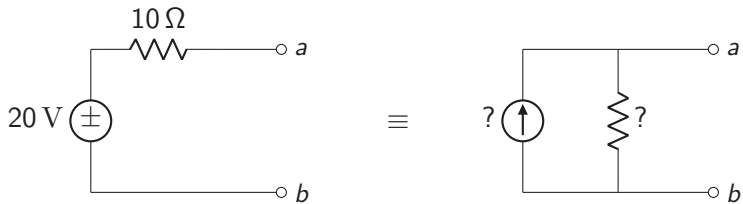
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

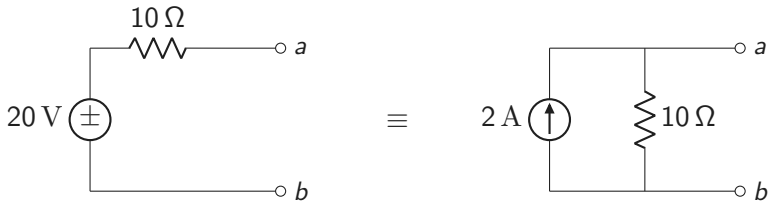


Example: a source transformation



Find the current source and the resistor.

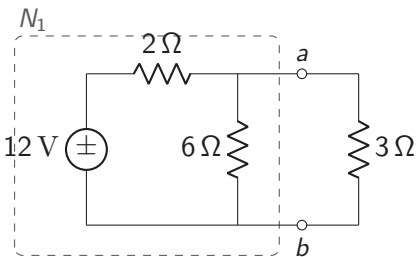
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

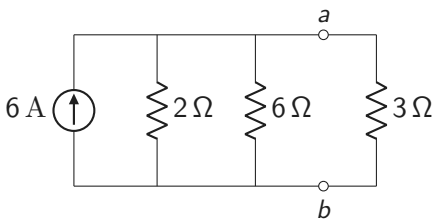
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

Example: the substitution example, by transformations



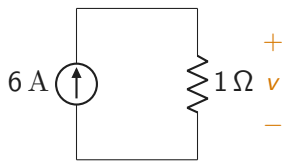
Find v again, with source transformations only.

Example: the substitution example, by transformations



$12\,\mathrm{V}$ with $2\,\Omega$ in series $\rightarrow$ $6\,\mathrm{A}$ with $2\,\Omega$ in parallel.

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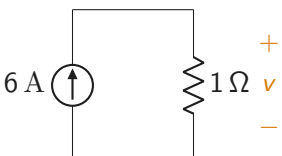


$12\,\mathrm{V}$ with $2\,\Omega$ in series $\rightarrow$ $6\,\mathrm{A}$ with $2\,\Omega$ in parallel.

$2\,\Omega$, $6\,\Omega$, $3\,\Omega$ in parallel:
$G_{eq} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1\,\mathrm{S}$, so $R_{eq} = 1\,\Omega$.

$v = 6\,\mathrm{A} \times 1\,\Omega = 6\,\mathrm{V}, \quad i = 6/3 = 2\,\mathrm{A}.$ ✓

Example: the substitution example, by transformations



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$v = 6\,\mathrm{A} \times 1\,\Omega = 6\,\mathrm{V}, \quad i = 6/3 = 2\,\mathrm{A}.$ ✓

Careful: inside the transformed one-port the values differ. The $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $6\,\mathrm{A}$ source carries $6\,\mathrm{A}$.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.

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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.

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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.